

CSE422: Artificial Intelligence

Project Report

Project:

Prediction of tree sterility

Submitted by

Group no: 02

Section: 19

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1. Introduction

This project aims to build an intelligent system that predicts tree sterility using machine learning techniques. Tree sterility—the inability to produce viable seeds or fruit—affects biodiversity, ecosystems, and agriculture. Motivated by environmental concerns and climate change, we analyze biological and environmental data to identify patterns linked to sterility. The process includes data preprocessing, exploratory analysis, and training classification models like Neural Networks, Decision Trees, and Logistic Regression. We compare model performance to find the most accurate predictor and highlight the key features influencing sterility.

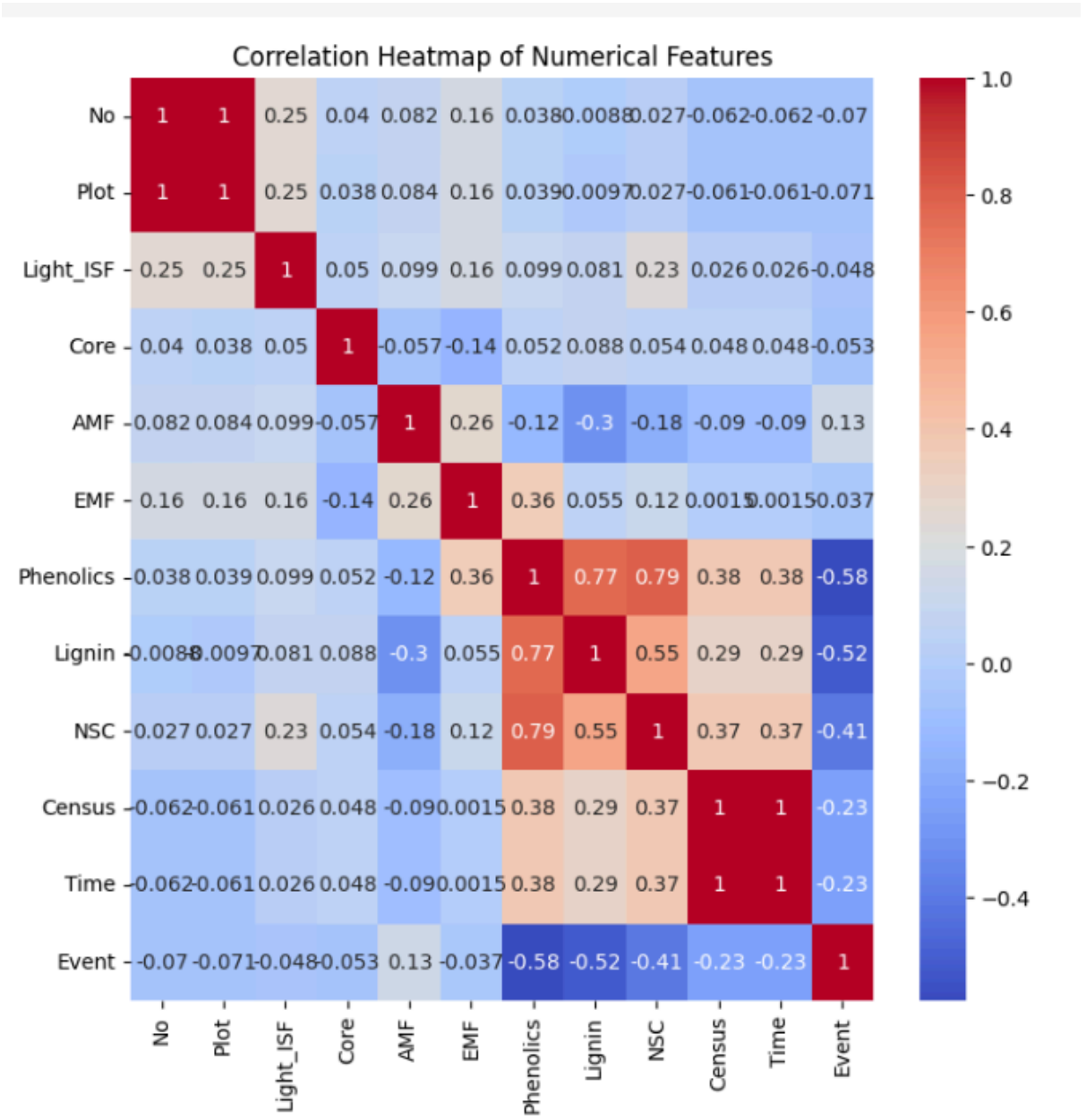
2. Dataset Description

- **Source:**

  https://drive.google.com/file/d/11C-tR5vXotRiA05yq44pubXbIXrzxVTS/view

- **Features:** 24
- **Classification/Regression:** It is a classification problem as the target column being predicted has two values: Sterile or Non-Sterile
- **Data points:** 2783 rows

- **Feature types:**
  - **Quantitative**: ['No', 'Plot', 'Light_ISF', 'Core', 'AMF', 'EMF', 'Phenolics', 'Lignin', 'NSC', 'Census', 'Time', 'Event']
  - **Categorical**: ['Subplot', 'Species', 'Light_Cat', 'Soil', 'Adult', 'Sterile', 'Conspecific', 'Myco', 'SoilMyco', 'PlantDate', 'Harvest', 'Alive']
- **Correlation of all features:**



- **What do you understand after the correlation test?**

Strong Positive Correlations (≥ ±0.75)

No & Plot: 0.998

→ Nearly identical; likely represent the same or sequential IDs.

Census & Time: 1.000

→ Perfectly correlated, likely duplicate or functionally identical.

Phenolics & NSC: 0.791

→ As phenolics increase, NSC (non-structural carbohydrates) also increases.

Phenolics & Lignin: 0.768

→ Strong positive correlation, not negative as previously interpreted.

Strong Negative Correlations (≤ -0.5)

Event & Phenolics: -0.576

Event & Lignin: -0.520

Event & NSC: -0.410 (Moderate but notable)

→ Events may correspond to decreased levels of these biochemical features.

Moderate Correlations (±0.3 to ±0.7)

Phenolics & EMF: 0.360

NSC & Lignin: 0.547

NSC & Phenolics: 0.791

Lignin & Phenolics: 0.768

Census & NSC: 0.368

Census & Phenolics: 0.380

→ These show meaningful but not extreme relationships.

Weak to Very Weak Correlations (±0.0 to ±0.3)

Core & Any Variable: mostly around ±0.05

Light_ISF with Most Variables: e.g., with Core (0.05), AMF (0.10)

AMF & EMF: 0.265

No/Plot & Almost Everything Else: mostly < ±0.1

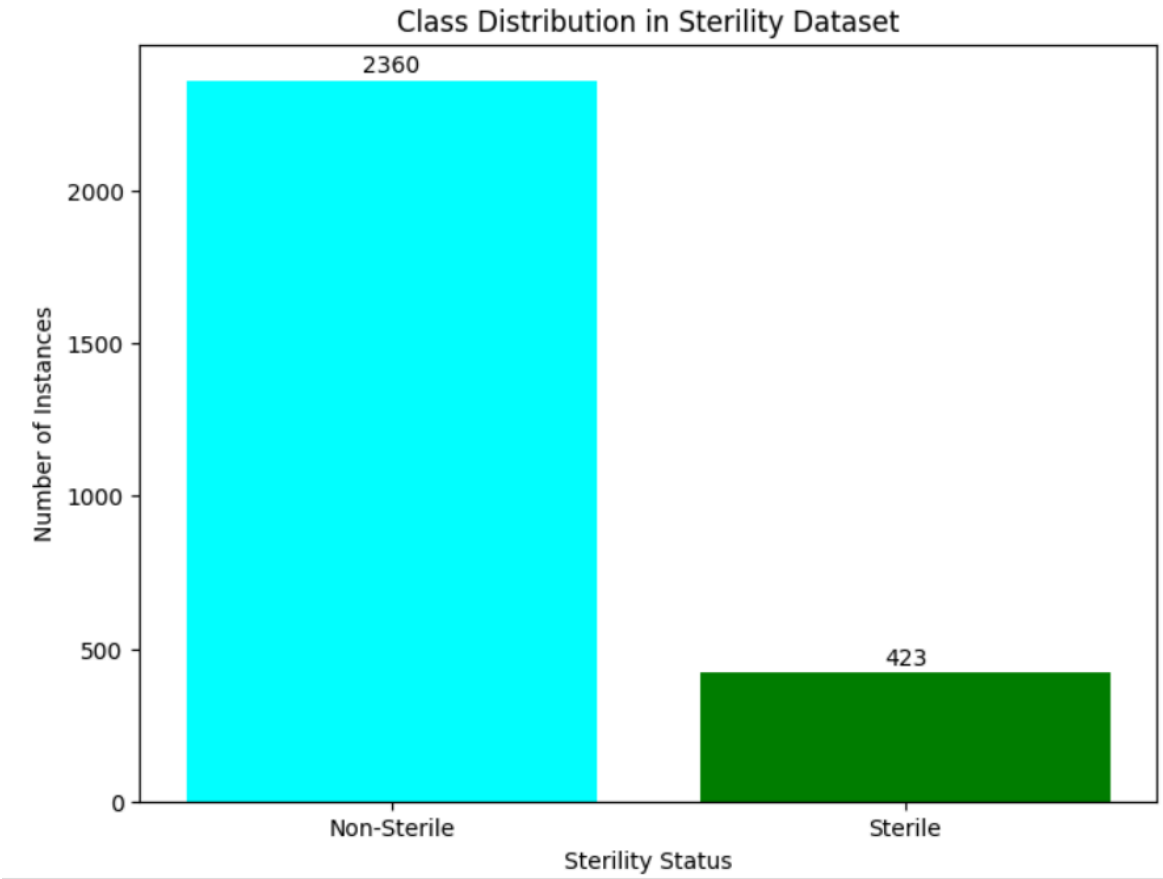
→ These likely don't contribute much to prediction on their own.

- **Imbalanced Dataset:**

  For the output feature, do all unique classes have an equal number of instances or not? The dataset is highly imbalanced:

  Non-Sterile: 2,360 instances

  Sterile: 423 instances



- **EDA Observation:**

  We analyzed the dataset to understand its structure and feature relationships. It includes 12 numeric and 12 categorical features. By checking unique value counts and variance, we identified how spread out each feature is and noticed that some, like Harvest and Alive, may be irrelevant. The correlation matrix and heatmap helped reveal how features influence each other, while density plots showed that No and Plot, as well as Time and Census, are nearly identical. We also observed an imbalance in the target variable, Sterile, which may affect model performance.

- **Final Observation:**

  Around 84.8% of instances consist of outcome value 0 and 15% of instances consist of outcome value 1.

3. Dataset Preprocessing

To prepare the dataset for modeling, we first removed features like Alive and Harvest because they contained many null or irrelevant values. Features such as EMF and Event also had missing data, but since they included important information, we chose not to drop them. Instead, we filled in the missing values using the average of existing values—a technique known as data imputation. Additionally, some features were in string format, which machine learning models can't process directly, so we converted them into numerical form through categorical encoding. Finally, we applied standard scaling to improve model accuracy, although we noted that if the dataset is imbalanced, scaling may sometimes introduce issues.

4. Dataset splitting

We performed Stratified data splitting (70-30) to ensure that both the training and testing sets maintain the same proportion of sterile and non-sterile trees as found in the original dataset.

5. Model Training and Testing

🔹 Logistic Regression:

The logistic regression model achieved perfect performance with an accuracy of 100%. The confusion matrix shows zero misclassifications, and the classification report confirms a precision, recall, and F1-score of 1.00 for both sterile and non-sterile classes.

🔹 Random Forest:

Similar to logistic regression, the random forest model also yielded flawless results with 100% accuracy. All predictions were correct, as indicated by the confusion matrix and the classification report showing perfect precision, recall, and F1-score.

🔹 K-Nearest Neighbors (K=5):

The KNN model achieved a slightly lower accuracy of 97.03%. It misclassified 28 samples in total—20 non-sterile trees as sterile and 8 sterile trees as non-sterile. Despite the few errors, it maintained strong precision and recall values of around 0.96–0.98.

🔹 Neural Network:

The neural network model performed identically to the logistic regression and random forest models, achieving a perfect 100% accuracy. The confusion matrix showed no errors, and the classification metrics were all at their maximum value of 1.00.

The 100% accuracy likely indicates that the dataset is either too simple, has highly separable features,

6. Model Selection or Comparison Analysis

As we have a classification problem we ran precision,recall comparison, confusion matrix, AUC score, ROC curve for each model.

🔹 **Model: Logistic Regression**

Accuracy: 1.0

Precision: 1.0

Recall: 1.0

F1 Score: 1.0

ROC AUC Score: 1.0

Classification Report:

	Precision	Recall	F1-Score	Support
Non-Sterile	1.00	1.00	1.00	708
Sterile	1.00	1.00	1.00	708

Accuracy			1.00	1416
Macro avg	1.00	1.00	1.00	1416
Weighted avg	1.00	1.00	1.00	1416

- **Model: Random Forest**

Accuracy: 1.0

Precision: 1.0

Recall: 1.0

F1 Score: 1.0

ROC AUC Score: 1.0

Classification Report:

	Precision	**Recall**	**F1-Score**	**Support**
Non-Sterile	1.00	1.00	1.00	708
Sterile	1.00	1.00	1.00	708
Accuracy			1.00	1416
Macro avg	1.00	1.00	1.00	1416
Weighted avg	1.00	1.00	1.00	1416

- **Model: KNN (K=5)**

Accuracy: 0.9675

Precision: 0.9572

Recall: 0.9788

F1 Score: 0.9679

ROC AUC Score: 0.9675

Classification Report:

	Precision	Recall	F1-Score	Support
Non-Sterile	0.98	0.96	0.97	708
Sterile	0.96	0.98	0.97	708
Accuracy			0.97	1416
Macro avg	0.97	0.97	0.97	1416
Weighted avg	0.97	0.97	0.97	1416

- **Model: Neural Network**

Accuracy: 1.0

Precision: 1.0

Recall: 1.0

F1 Score: 1.0

ROC AUC Score: 1.0

Classification Report:

	Precision	Recall	F1-Score	Support
Non-Sterile	1.00	1.00	1.00	708
Sterile	1.00	1.00	1.00	708
Accuracy			1.00	1416

Macro avg	1.00	1.00	1.00	1416
Weighted avg	1.00	1.00	1.00	1416

8. Conclusion

The high accuracy achieved by the neural network model suggests that tree sterility can be effectively predicted using features like species, soil, and light, indicating strong biological and environmental influences. However, since the dataset is imbalanced, this high accuracy may be misleading and could be due to overfitting or data leakage. The imbalance caused the model to perform unrealistically well, sometimes reaching 100% accuracy, especially after feature scaling. To address this, we also tested the models on unscaled data to better evaluate their true performance and reliability.